

EX NO: 2a	GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB
DATE:	

```
%=====
% Experiment: Frequency Modulation (FM)
%=====
```

```
clc;
clear;
close all;
```

```
%% Input Parameters
```

```
Am = 1;      % Message signal amplitude
Ac = 1;      % Carrier signal amplitude
```

```
fm = 5;      % Message frequency (Hz)
fc = 50;     % Carrier frequency (Hz)
```

```
beta = 5;    % Modulation index
kp = 5;      % Phase sensitivity (not used directly)
```

```
fs = 1000;   % Sampling frequency
```

```
%% Time Vector
t = 0:1/fs:1;
```

```
%% Message Signal
m = Am*cos(2*pi*fm*t);
```

```
%% Carrier Signal
c = Ac*cos(2*pi*fc*t);
```

```
%% Frequency Modulated Signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
```

```
%% Frequency Spectrum
N = length(fm_sig);
f = linspace(-fs/2, fs/2, N);
```

```
FM_FFT = abs(fftshift(fft(fm_sig)))/N;
```

```
%% Display Results
```

```
fprintf('n===== FM MODULATION RESULTS =====\n');
fprintf('Message Amplitude   = %.2f V\n', Am);
fprintf('Carrier Amplitude     = %.2f V\n', Ac);
fprintf('Message Frequency      = %d Hz\n', fm);
fprintf('Carrier Frequency      = %d Hz\n', fc);
fprintf('Sampling Frequency     = %d Hz\n', fs);
fprintf('Modulation Index (β)   = %.2f\n', beta);
```

```

fprintf('Frequency Deviation   = %.2f Hz\n', beta*fm);
fprintf('Bandwidth (Carson Rule)= %.2f Hz\n', 2*(beta+1)*fm);
fprintf('===== \n');

```

```

%% Plot Results
figure('Name','Frequency Modulation (FM)','NumberTitle','off');

```

```

% Message Signal
subplot(4,1,1);
plot(t,m,'b','LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```

```

% Carrier Signal
subplot(4,1,2);
plot(t,c,'r','LineWidth',1.5);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```

```

% FM Signal
subplot(4,1,3);
plot(t,fm_sig,'k','LineWidth',1.5);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```

```

% FM Spectrum
subplot(4,1,4);
plot(f,FM_FFT,'m','LineWidth',1.5);
xlim([fc-50 fc+50]);
title('FM Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

```

Frequency Modulation (FM)



FIGURE

FORMAT

TOOLS

LAYOUT

